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MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M5A1 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers metal parts assembly for Catapult, Aircraft Ejection Seat, M5A1. Assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in air-crew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect, on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for
MIL–I–45208 —Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–109 —Quality Assurance Terms and Definitions
MIL–STD–413 —Visual Inspection Guide for Rubber O-Rings
MIL–STD–639 —Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

MUNITIONS COMMAND

F8593687 —Catapult, Aircraft Ejection Seat, M5A1, Metal Parts Assembly
F8797377 —Ammunition, Container, Fiber M299 for Catapult, Aircraft Ejection Seat, M5A1
F8797378 —Box, Packing, Ammunition, for Catapult, Aircraft Ejection Seat, M5A1
IEL–8593687 —Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat, M5A1, Metal Parts Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with all the requirements specified on the applicable drawings and specifications.

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

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3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties. Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure. The components listed below shall withstand the corresponding internal pressure as indicated in table I for a period of 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

TABLE I

Component	Hydrostatic pressure (psi, minimum)
Outside tube	3400
Telescoping tube	2500
Inside tube	3100

3.6 Shear pin strength. After application of protective finish, the pin shall shear when a load of 400 plus or minus 50 pounds is applied at the steady rate of .25 inches per minute.

3.7 Structural strength. The metal parts shall be capable of withstanding each test load for 15 seconds, minimum, as indicated in table II. The maximum allowable change between mounting points, resulting from each test load application shall not exceed the values indicated in table II. A preload of 500 pounds shall be utilized prior to each test load application.

3.8 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

TABLE II

Test loads (min.)	Change between mounting points (max.)
1. 2000 pounds in tension	.005 inches
2. 2000 pounds in tension	.001 inches
3. 8000 pounds in compression	.005 inches
4. 8000 pounds in compression	.001 inches

3.8.1 *Torque requirements.* Torque requirements shall be as specified in Table III. No loosening or deformation shall occur upon application of the breakaway torque.

3.8.1.1 *Bottoming.* The bottoming torque between block and inside tube and between stop ring and trunnion shall not exceed the running torque by more than 50 inch pounds.

TABLE III

Components	*A* Run-on (inch pounds)	*B* Seating or bottoming (including A) (inch pounds)	*C* Breakaway (inch pounds)
Cap and Outside Tube.	75 Min.	500+50	400 Min.
Trunnion and Outside Tube.	75 Min.	550+50	400 Min.
Block and Inside Tube	75 Min.	.....	.....
Stop Ring and Trunnion.	75 Min.	.....	.....

3.9 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.10 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.11.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 *Bur.* No part shall have a bur which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies and 10 sets of parts and 50 shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4, except as provided in MIL–I–45208. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Presentation of lots.* The product shall be presented in accordance with MIL–STD–105.

4.3.1.2 *Size of lot.* The shipping lots of metal parts assemblies shall consist of 614 catapults, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables IV and V. For an individual defect (single characteristic) the criteria in table IV shall be used. For total defects within a major or minor group the criteria in table V shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL–STD–413. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as bur and foreign matter.

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TABLE IV

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1— 300	100% examination				
301— 1,300	110	1	2	2	3
1,301— 3,200	150	2	3	3	4
3,201— 8,000	225	3	4	4	5
8,001—22,000	300	4	5	5	6

TABLE V. Total allowable defects per sample size

Sample size for major defects	110	150	225	300
No. of Defects in Class				
2— 3	1	2	3	4
4— 7	2	3	4	5
8—19	3	4	5	7
20—43	4	5	8	10
Over—43	6	8	11	14

Sample size for major defects	110	150	225	300
No. of Defects in Class				
2— 3	2	3	4	5
4— 7	3	4	5	7
8—16	4	5	8	10
17—48	6	8	11	14
over—48	8	11	17	20

4.3.2.1 *Latch, Dwg. C8593644, covering a detail of Dwg. F8593687.*

Defect	*Method of inspection*
Critical	
None defined	
Major	
101. Diameter of firing pin relief	Gage
102. Length to inside surface, small end	Gage
103. Length of cut out	Gage
104. Width at small end	Gage
105. Width at cut out	Gage
106. Thickness at datum location	Gage
107. Location of firing pin relief	Gage
108. Angle at large end	Gage
109. Metal defective	Visual
110. Protective finish inadequate or defective	Visual
111. Surface finish of base, chamfer and face, improper	Visual
112. Bur or sharp corner	Visual
113. Foreign matter	Visual
Minor	
201. Total length, max.	Gage
202. Width at large end	Gage
203. Thickness, max.	Gage
204. Thickness at firing pin relief, max.	Gage

4.3.2.2 *Block, Dwg. F8593648, covering a detail of Dwg. F8593687.*

Defect	*Method of inspection*
Critical	
None defined	
Major	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Pitch diameter of pipe thread	Gage
104. Minor diameter of pipe thread, max.	Gage
105. Concentricity of counterbore with small cavity	Gage
106. Concentricity of pitch diameter of thread with counterbore	Gage
107. Concentricity of pitch diameter of thread with body	Gage
108. Concentricity of large cavity with counterbore	Gage
109. Concentricity of body shoulder with body	Gage
110. Concentricity of flange with body shoulder	Gage
111. Symmetry of length of slot with body shoulder	Gage
112. Symmetry of thickness of rectangular section with body shoulder	Gage
113. Symmetry of width of rectangular section with body shoulder	Gage
114. Perpendicularity of hole with rectangular section	Gage
115. True position of pin hole with small cavity	Gage
116. True position of hole in rectangular section with body shoulder	Gage
117. True position of screw hole with counterbore and base of large cavity	Gage
118. Diameter of mounting hole in rectangular section	Gage
119. Diameter of small cavity	Gage
120. Diameter of counterbore, max.	Gage
121. Depth of drill cavity, max.	Gage
122. Length to flange, max.	Gage
123. Length of effective thread, min.	Gage
124. Width of rectangular section, max.	Gage
125. Diameter of thread undercut, max., if applicable	Measure
126. Depth to bottom of drilled hole, max.	Measure
127. Location of pipe hole, max.	Measure
128. Thread damaged or not full	Visual
129. Metal defective	Visual

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Defect	Method of inspection
130. Protective finish inadequate or defective	Visual
131. Surface finish of small cavity, body shoulder and body slots, improper	Visual
132. Bur	Visual
133. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, min.	Gage
202. Diameter of large cavity	Gage
203. Diameter of counterbore, min.	Gage
204. Diameter of body shoulder	Gage
205. Diameter of flange, max.	Gage
206. Diameter of pin hole in body	Gage
207. Diameter of screw hole, min.	Gage
208. Depth of large cavity	Gage
209. Length from flange to base of large cavity	Gage
210. Length of slot in body	Gage
211. Width of slot in body	Gage
212. Location of slot to base of large cavity	Gage
213. Location of screw hole	Gage
214. Location of hole in rectangular section	Gage
215. Location of pin hole in body	Gage
216. Depth of counterbore	Measure
217. Profile at ends of rectangular section	Measure

4.3.2.3 *Pin, Shear, Dwg. B8593654, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Bur	Visual
106. Foreign matter	Visual
Minor	
201. Total length, max.	Gage

4.3.2.4 *Plug, Dwg. C8593656, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
1. Diameter of boss hole, min.	Gage
Major	
101. Concentricity of boss hole with cavity	Gage
102. Concentricity of flange with cavity	Gage
103. Concentricity of body with flange	Gage
104. Symmetry of base slot with cavity	Gage
105. True position of pitch diameter of set screw thread with flange including perpendicularity with slot	Gage
106. True position of pin hole with cavity and slot	Gage
107. Diameter of cavity	Gage
108. Diameter of flange, min.	Gage
109. Depth of cavity, min.	Gage
110. Depth of slot	Gage
111. Total length	Gage
112. Height of boss	Gage
113. Chamfer on boss hole, improper	Visual
114. Metal defective	Visual
115. Protective finish inadequate or defective	Visual
116. Surface finish of cavity and slot, improper	Visual
117. Bur or sharp corner	Visual
118. Foreign matter	Visual
Minor	
201. Pitch diameter of thread	Gage
202. Minor diameter of thread, max.	Gage
203. Diameter of flange, max.	Gage
204. Diameter of body	Gage
205. Diameter of pin hole, min.	Gage
206. Length of effective thread, min.	Gage
207. Width of slot	Gage
208. Location of pin hole	Gage
209. Location of threaded hole	Gage

4.3.2.5 *Pin, Firing, Dwg. C8593657, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
1. Diameter of striker, max.	Gage
Major	
101. Concentricity of striker with striker shoulder	Gage
102. Concentricity of flange with striker shoulder	Gage
103. Concentricity of groove with flange	Gage
104. Symmetry of forward datum location with striker shoulder	Gage
105. Symmetry of width of body with striker shoulder	Gage
106. Symmetry of thickness of body with striker shoulder	Gage
107. True position of pin hole with striker shoulder and thickness of body	Gage
108. Diameter of striker shoulder, min.	Gage
109. Diameter of flange, min.	Gage
110. Diameter of groove	Gage
111. Depth of hole in striker should, max.	Gage
112. Length of striker, min.	Gage
113. Width of body, max.	Gage
114. Width of groove	Measure
115. Profile of relief groove on striker shoulder	Visual
116. Metal defective	Visual

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Defect	Method of inspection
117. Protective finish inadequate or defective	Visual
118. Surface finish of body, flange and striker shoulder improper	Visual
119. Bur or sharp corner, except as permitted	Visual
120. Foreign matter	Visual
Minor	
201. Diameter of striker shoulder, max.	Gage
202. Diameter of flange, max.	Gage
203. Diameter of hole in striker shoulder	Gage
204. Depth of hole in striker shoulder, min.	Gage
205. Width at forward datum location	Gage
206. Width at rear datum location	Gage
207. Location of hole	Gage
208. Surface variation on rear face of striker shoulder	Gage

4.3.2.6 *Seal, Plain, Dwg. B8593658, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Diameter of wire, max. (after plating)	Measure
102. Number of coils, incorrect	Visual
103. Locational variance of ends, max.	Visual
104. Metal defective	Visual
105. Protective finish, inadequate or defective	Visual
106. Bur or sharp edge	Visual
107. Foreign matter	Visual
Minor	
201. Diameter, outside	Measure
202. Direction of turn, incorrect	Visual

4.3.2.7 *Seal, Fixed, Dwg. B8593660, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Diameter of wire, max. (after plating)	Gage
102. Length of tang	Gage
103. Locational variance of ends, max.	Visual
104. Number of coils, incorrect	Visual
105. Radius of tang bend, max.	Visual
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Bur or sharp edge	Visual
109. Foreign matter	Visual
Minor	
201. Diameter, inside	Gage
202. Direction of turn incorrect	Visual

4.3.2.8 *Ring, Stop, Dwg. C8593642–43, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of bore with pitch diameter of thread	Gage
104. Concentricity of counterbore with bore	Gage
105. Symmetry of width across flats with flange	Gage
106. Diameter of bore, min.	Gage
107. Diameter of thread undercut, min.	Gage
108. Length to flange	Gage
109. Length to counterbore	Gage
110. Thread damaged or not full	Visual
111. Pellet missing	Visual
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish of angle of bore and bore, improper	Visual
115. Bur or sharp corner	Visual
116. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage
202. Diameter of counterbore	Gage
203. Diameter of flange	Gage
204. Diameter of bore, min.	Gage
205. Total length	Gage

4.3.2.9 *Tube, Outside, Dwg. D8593689–91, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Pitch diameter of thread, min. (both ends)	Gage
102. Major diameter of thread, min. (both ends)	Gage
103. Concentricity of pitch diameter of thread with inside diameter at datum locations	Gage
104. Diameter, inside	Gage
105. Total length, min.	Gage
106. Length of effective thread, min.	Gage
107. Thickness of wall, min.	Gage
108. Straightness	Gage
109. Pellet missing	Visual
110. Thread damaged or not full	Visual
111. Protective finish inadequate or defective	Visual
112. Bur	Visual
113. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage

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Defect	Method of inspection
202. Diameter of body, max.	Gage
203. Diameter, inside, min.	Gage
204. Total length, max.	Gage
205. Length of thread, max.	Gage

4.3.2.10 *Tube, Telescoping, Dwg. D8593693, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Diameter of body	Gage
102. Diameter of shoulder	Gage
103. Diameter, inside	Gage
104. Total length, min.	Gage
104. Total length, min.	Gage
105. Thickness of wall variation, max.	Gage
106. Straightness	Gage
107. Diameter of holes in shoulder	Measure
108. Number of holes in shoulder, incorrect	Visual
109. Metal defective	Visual
110. Protective finish inadequate or defective	Visual
111. Surface finish of shoulder, improper	Visual
112. Bur	Visual
113. Foreign matter	Visual
Minor	
201. Total length, max.	Gage

4.3.2.11 *Tube, Inside, Dwg. D8593695, –97, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with shoulder at datum location	Gage
104. Concentricity of pitch diameter of thread with bore at datum location	Gage
105. Diameter of shoulders	Gage
106. Depth of relief of spiral groove, min.	Gage
107. Total length, max.	Gage
108. Thickness of wall, min.	Gage
109. Length of effective thread, min.	Gage
110. Straightness	Gage
111. Pellet missing	Visual
112. Metal defective	Visual
113. Thread damaged or not full	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish of body at ends, improper	Visual
116. Bur	Visual
117. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage
202. Depth of spiral groove	Gage
203. Total length, max.	Gage
204. Length of relief of spiral groove, max.	Measure
205. Width of relief of spiral groove, max.	Measure
206. Location of spiral groove	Measure
207. Direction of turn of spiral groove, incorrect	Visual
208. Number of turn of spiral groove, incorrect	Visual

4.3.2.12 *Cap, Dwg. C8593699, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of outside diameter with pitch diameter thread	Gage
104. Diameter of thread undercut, max.	Gage
105. Diameter of body, min.	Gage
106. Width of thread undercut, max.	Gage
107. Thickness of base	Gage
108. Thread damaged or not full	Visual
109. Metal defective	Visual
110. Protective finish inadequate or defective	Visual
111. Bur or sharp corner	Visual
112. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, min.	Gage

4.3.2.13 *Trunnion, Dwg. D8593700, covering a detail of Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Pitch diameter of small thread, max.	Gage
102. Minor diameter of small thread, max.	Gage
103. Pitch diameter of large thread, max.	Gage
104. Minor diameter of large thread, max.	Gage
105. Concentricity of pitch diameter of large thread with inside diameter	Gage
106. Concentricity of inside diameter with pitch diameter of small thread	Gage
107. Symmetry of lugs	Gage
108. Symmetry of width across flats at root of lugs with inside diameter	Gage
109. True position of lug with inside diameter	Gage
110. Concentricity of lugs	Gage

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Defect	Method of inspection
111. Diameter, inside	Gage
112. Diameter of lugs	Gage
113. Diameter of small thread undercut, max.	Gage
114. Diameter of large thread undercut, max.	Gage
115. Depth of small thread cavity, min.	Gage
116. Depth of large thread cavity, min.	Gage
117. Depth of relief of spiral groove, min.	Gage
118. Length to center of lug	Gage
119. Length of effective thread, min.	Gage
120. Width across flats at root of lugs	Gage
121. Height of lug	Measure
122. Thread damaged or not full	Visual
123. Metal defective	Visual
124. Protective finish inadequate or defective	Visual
125. Surface finish of inside diameter, improper	Visual
126. Bur	Visual
127. Foreign matter	Visual
Minor	
201. Pitch diameter of small thread, min.	Gage
202. Pitch diameter of large thread, min.	Gage
203. Depth of spiral groove	Gage
204. Width of relief of spiral groove	Gage
205. Location of groove at small end	Gage
206. Diameter of tang hole at spiral groove	Measure
207. Length of relief of spiral groove	Measure
208. Location of spiral groove	Measure
209. Symmetry of tang hole with spiral groove	Visual
210. Direction of turn of spiral groove, incorrect	Visual
211. Number of turns of spiral groove incorrect	Visual

4.3.2.14 *Metal Parts Assembly, Dwg. F8593687.*

Defect	Method of inspection
Critical	
None defined	
Major	
101. Any part missing	Visual
102. Improper assembly	Manual/Visual
103. Thread damaged or not full	Visual
104. Bur	Visual
105. Foreign matter	Visual
Minor	
None defined	

4.3.2.15 *Ammunition, Container, Fiber, Dwg. F8797377.*

Defect	Method of inspection
Critical	
None defined	
Major	
None	
Minor	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damage through all impregnated layers	Visual

4.3.2.16 *Box, Packing, Ammunition, Dwg. F8797378.*

Defect	Method of inspection
Critical	
None defined	
Major	
None defined	
Minor	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4 except as provided in MIL–I–45208.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot of components shall be tested to determine compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each outside, telescoping and inside tube shall be subjected to an internal hydrostatic pressure. Failure of any tube to comply with the requirements of 3.5 shall be cause for rejection of the tube.

4.3.3.3 *Shear pin strength.* Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.3.4 *Structural strength.* Fifteen complete assemblies shall be selected and sub-

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jected to the structural strength test. Failure of any assembly to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.3.3.5 *Breakaway torque.* A breakaway torque shall be applied to each assembly threaded joint as listed in table III. Failure of any assembly to comply with the requirements of column C table III shall be cause for rejection of the assembly.

4.3.3.6 *Protective finish.* The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawings.

4.3.3.7 *Residual magnetism.* Each metal parts assembly shall be tested for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593687, except as provided in MIL–I–45208.

4.4 Test methods and procedures.

4.4.1 *Hardness.* Prior to the application of protective finish, a Hardness Tester shall be used to determine compliance with the hardness requirements specified on the detail drawing.

4.4.2 *Hydrostatic pressure.* Prior to the application of protective finish, the outside, telescoping and inside tubes shall be mounted in the device and subjected to the internal hydrostatic pressure. The device used shall internally seal the tube within 1½ inches from each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 *Shear pin strength.* After application of protective finish, the pin shall be subjected to the shear strength test in accordance with the requirements of 3.6. The test shall be performed in the fixture in conjunction with the tensile testing machine.

4.4.4 *Structural strength.* The catapult shall be placed in a tensile testing machine and subjected to the specified tension and compression loads for determination of compliance with the requirements of 3.7. The test shall be performed as follows:

(a) Apply a preload of 500 pounds in tension and adjust the dial indicator to zero inches.

(b) Apply the load in tension so as to move the mounting points away from each other.

(c) Release the load in tension to 500 pounds in tension and record the dial indicator reading. Repeat (b) and (c).

(d) Apply a preload of 500 pounds in compression and adjust the dial indicator to zero inches.

(e) Apply the load in compression load so as to move the mounting points toward each other.

(f) Release the compression load to 500 pounds in compression and record the dial indicator reading. Repeat (e) and (f).

4.4.5 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.4.6 *Residual magnetism.* The catapult shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the catapult 6 inches from and at the same height as an approved compass, in a north-south horizontal position (trunnion end at south) with the compass set in an east-west heading as shown in Figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawing F8797377 and F8797378.

5.2 Data cards. Data card information shall be as specified in MIL–A–2550.

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6. NOTES

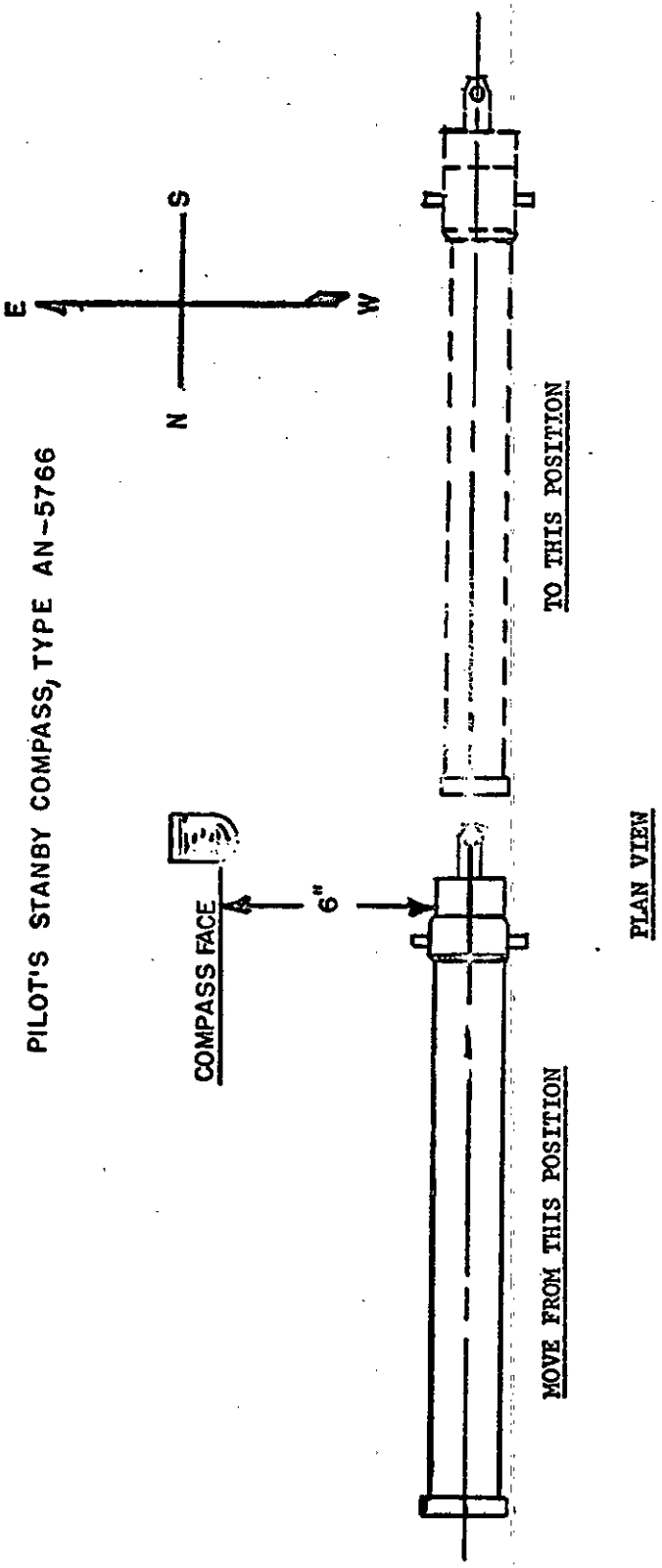
6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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NOTE:-

1. CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. *Layout for determination of residual magnetism in catapult, aircraft ejection seat, M5A1.*

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